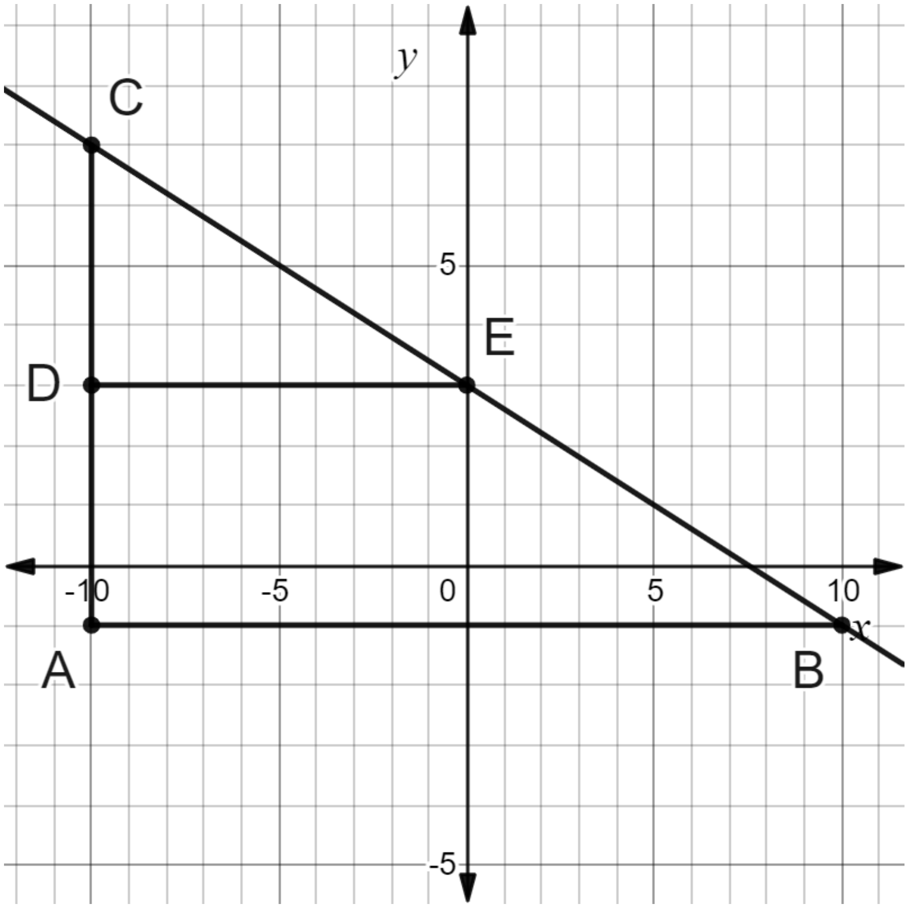
Grade 8
Unit 6 Assessment

Name______________________
Date__________________
Period__________

1. Similar triangles CDE and CAB are shown on the coordinate grid.



a. Complete the statement.

The ratios

- ○ $\frac{DE}{AC}$
- ○ $\frac{AB}{CB}$
- ○ $\frac{AC}{AB}$
- ○ $\frac{CD}{CE}$

and

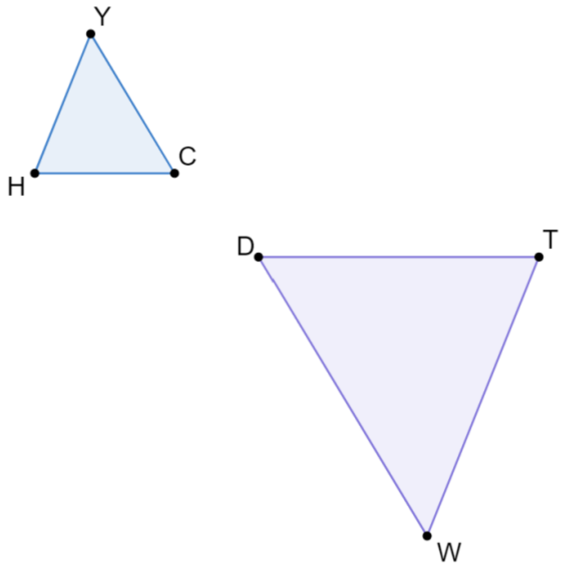
- ○ $\frac{CD}{DE}$
- ○ $\frac{DE}{CE}$
- ○ $\frac{AB}{EB}$
- ○ $\frac{CE}{CB}$

are equivalent to the slope of the line CB.

b. What is the slope of the line that passes through points C, E, and B?

Slope	$-\frac{4}{10}$ or equivalent

2. Similar triangles YHC and WTD and a table of some of the side lengths of the triangles are shown.

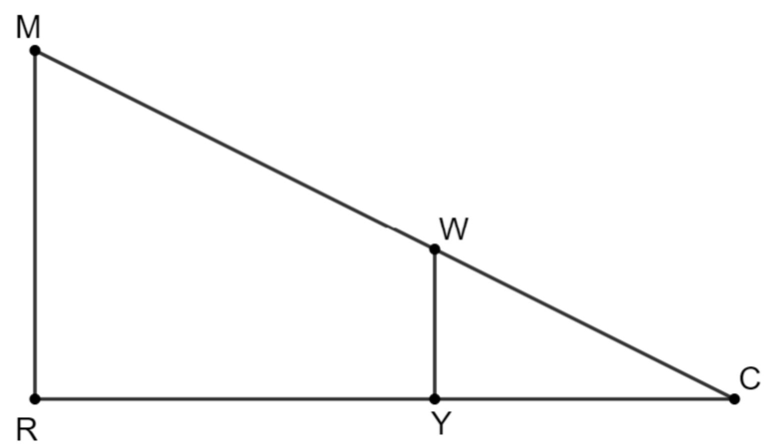


Side	Length
HY	54
TD	150
HC	50
TW	162
CY	58

Determine the constant of proportionality for triangles YHC and WTD.

$\dfrac{\textbf{corresponding side of } \Delta YHC}{\textbf{corresponding side of } \Delta WTD}$	$\frac{50}{150}$, $\frac{54}{162}$, or equivalent

3. Similar triangles MRC and WYC are shown where $MR = 14$, $CR = 28$, $WY = 6$, and $MW = 17.88$.

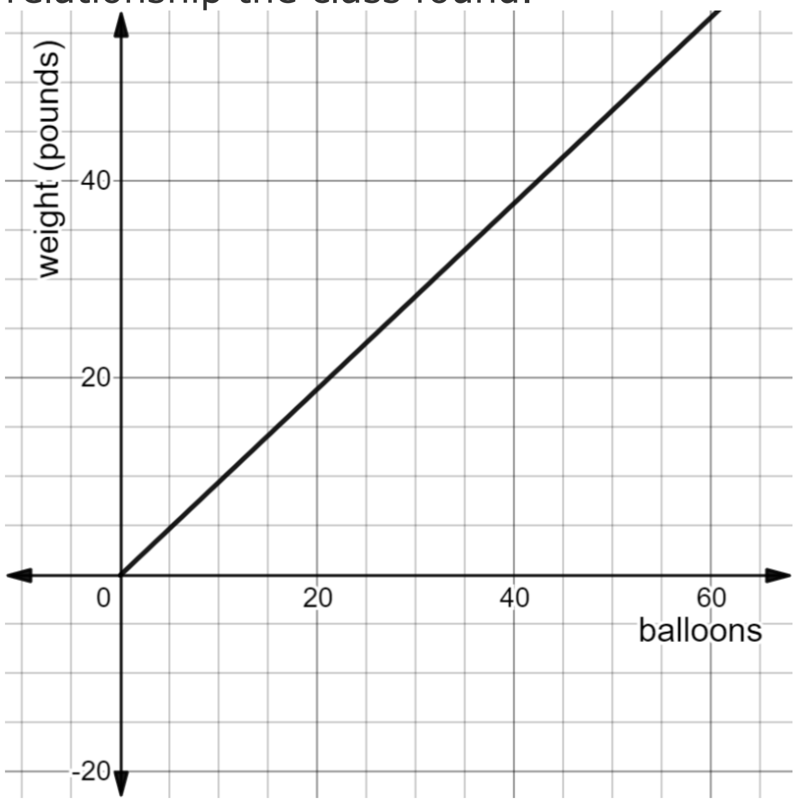


Determine the length of YC. Show your work.

Work

YC	12

4. A physics class did an experiment on how much weight helium balloons that are one-foot in diameter can lift. The graph models the linear relationship the class found.



a. Identify which variable is the independent variable and which is the dependent variable. Then, write the letter you want to use to represent each.

Independent Variable	number of balloons	**Letter**
		Answers will vary.
Dependent Variable	weight in pounds	**Letter**
		Answers will vary.

b. During the experiment the class found that that 54 helium balloons could lift 51 pounds. Write the two-variable equation, using the letters you indicated above, that models the relationship between the number of helium balloons and the weight, in pounds.

Equation	$y = \frac{51}{54}x$

**Variables used in equation should match letters in part a. Answer key uses x for the independent variable and y for the dependent.*

5. Allyson planted bamboo in her yard. Allyson measured the height, in inches (in.), of the bamboo. She found that the relationship between time and the height of the bamboo was linear. Her measurements are shown.

Time (hours)	24	30	36
Bamboo Height (in.)	42	51	60

a. Complete the statement.

The relationship between time and the height of the bamboo
- ○ is
- ○ is not

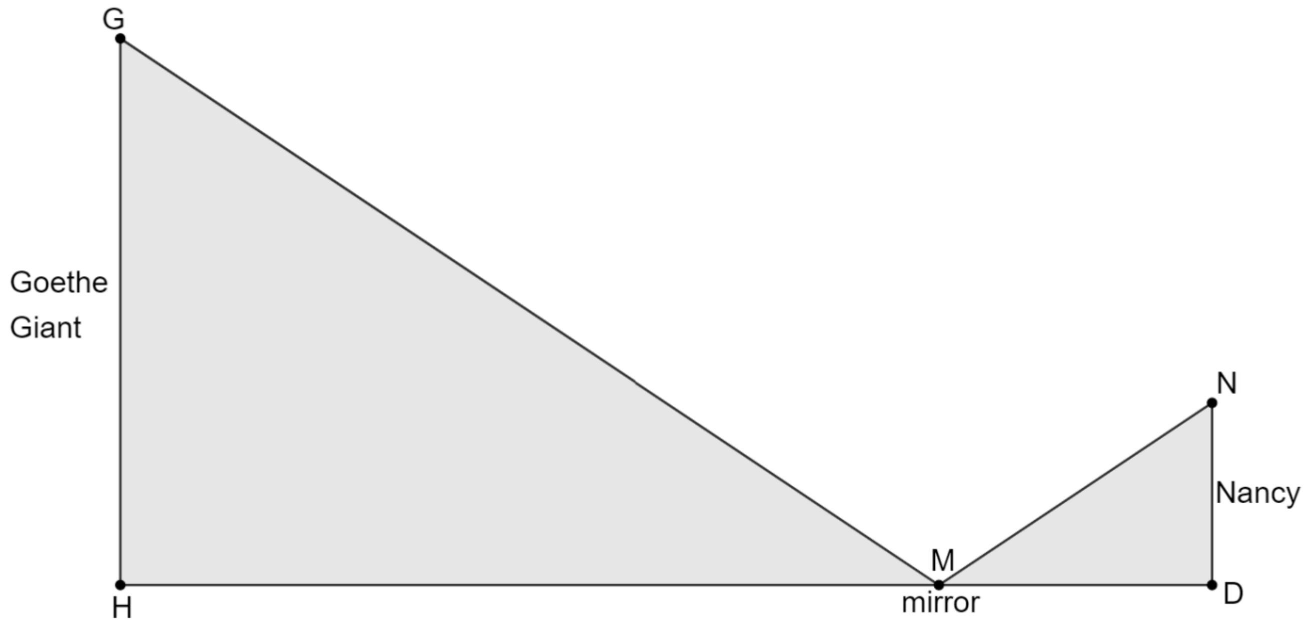
proportional because
- ○ time
- ○ the ratio $\frac{y}{x}$
- ○ the bamboo height

- ○ increases by the same amount
- ○ is not the same

for all of Allyson's measurements.

b. Justify the statement showing your work or explaining how you determined your answer.
Answers will vary.

6. The Goethe Giant is a Bald Cypress tree in Levy County's Goethe State Forest. The tree is 115 feet (ft) tall. Nancy stands 2 ft away from a mirror so she can see the top of the Goethe Giant. Nancy is 5 ft tall.

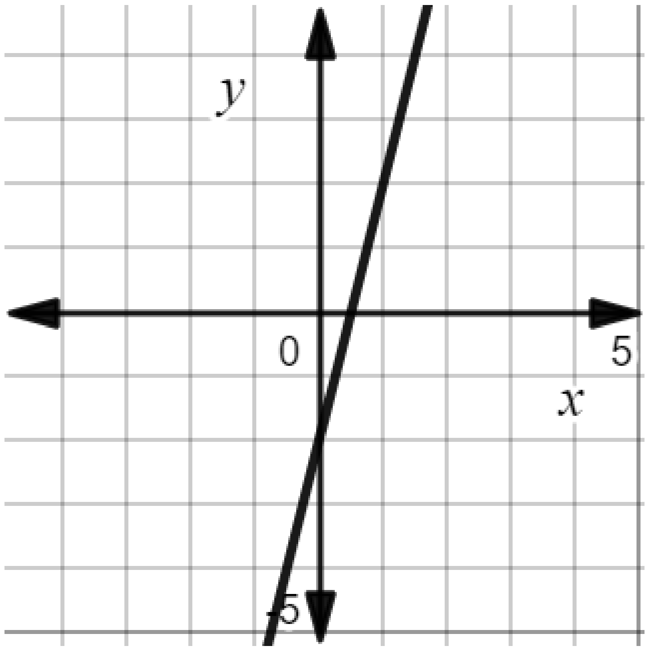


Determine $\overline{HM}$, the distance, in ft, the mirror is from the Goethe Giant. Show your work.

work

$\boldsymbol{HM}$	46

Determine if each linear relationship is proportional.

<table>
<tr><th></th><th>Relationship</th><th>Proportional or Not Proportional?</th></tr>
<tr><td>7.</td><td>$y = 5\frac{1}{2}x$</td><td>○ proportional
○ not proportional</td></tr>
<tr><td>8.</td><td>
</td><td>○ proportional
○ not proportional</td></tr>
<tr><td>9.</td><td>$y = \frac{6}{7}x + 8$</td><td>○ proportional
○ not proportional</td></tr>
<tr><td>10.</td><td><table>
<tr><th>x</th><th>y</th></tr>
<tr><td>-2</td><td>-5</td></tr>
<tr><td>7</td><td>17.5</td></tr>
<tr><td>10</td><td>25</td></tr>
<tr><td>13</td><td>32.5</td></tr>
</table></td><td>○ proportional
○ not proportional</td></tr>
</table>